ECS_TASK

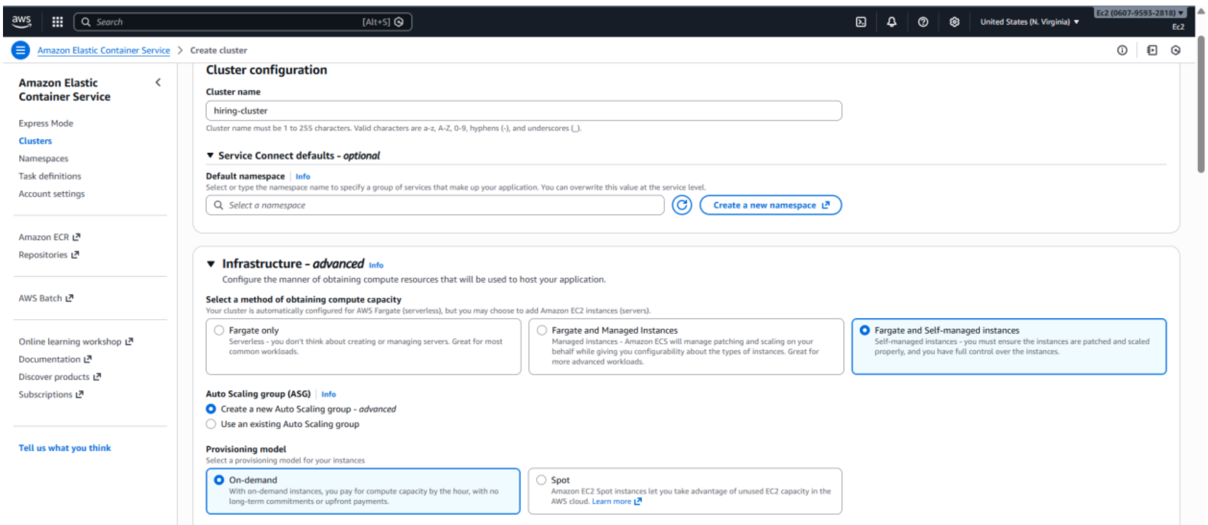
Task:

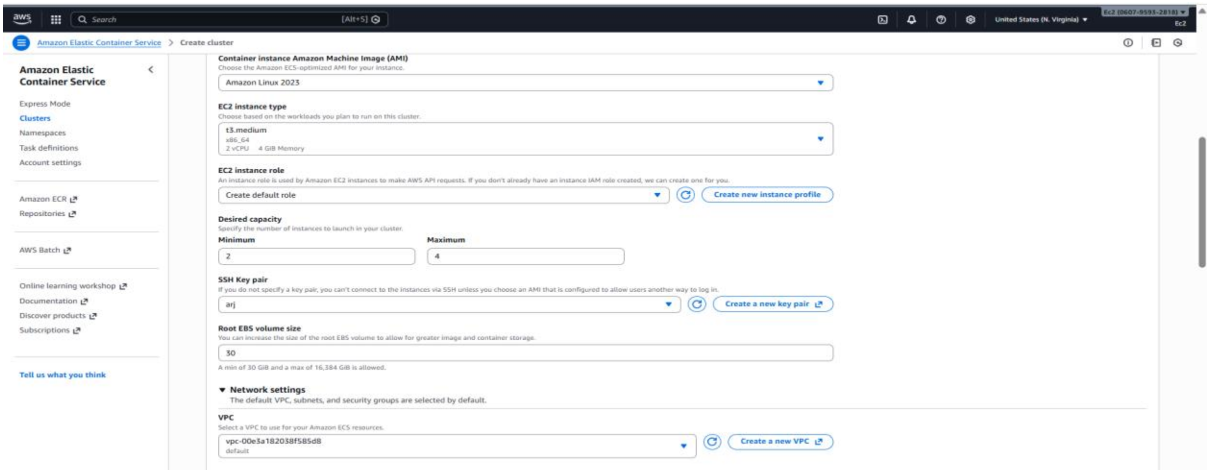
- **Setup a highly available ecs cluster with load balancer and dynamic port mapping.**
  NOTE:
  Your cluster should maintain atleast 6 tasks at any point of time and should be highly availble across multiple AZ's.
  Use the below image from deployment:
  sabair0509/hiring-app:works

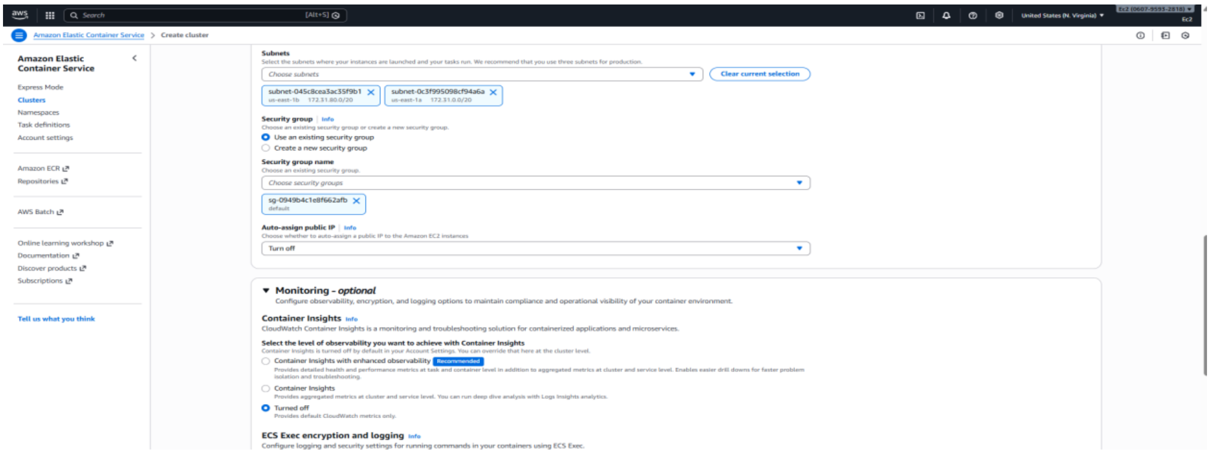
PHASE 1: Create ECS Cluster (EC2 + ASG)

1 On this screen set:

- Provisioning: **On-demand**
- Instance type: **t3.medium** (recommended)
- Instance role: **Create default role**
- Desired capacity: **2**
- Minimum: **2**
- Maximum: **4**
- Select **2 subnets in different AZ**
- Select key pair



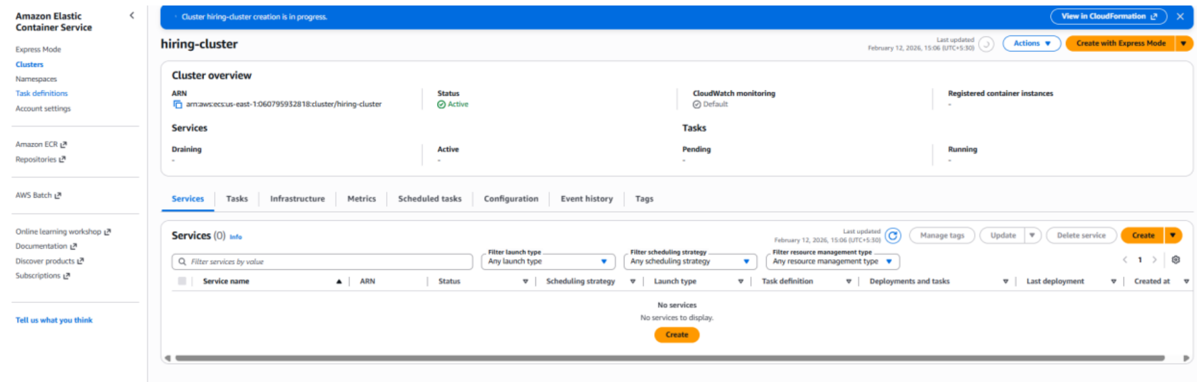


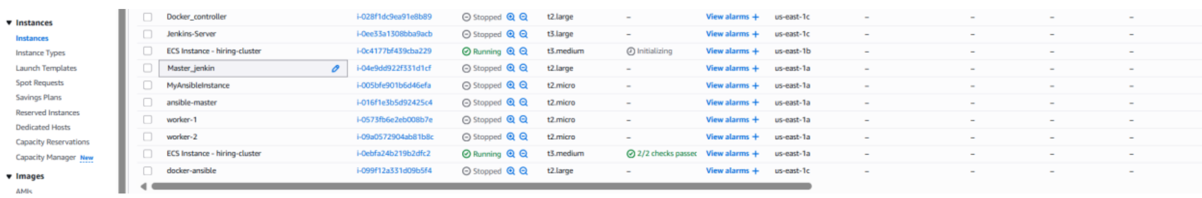


Now click:

✅ **Create cluster**

Wait 5-10 mins.



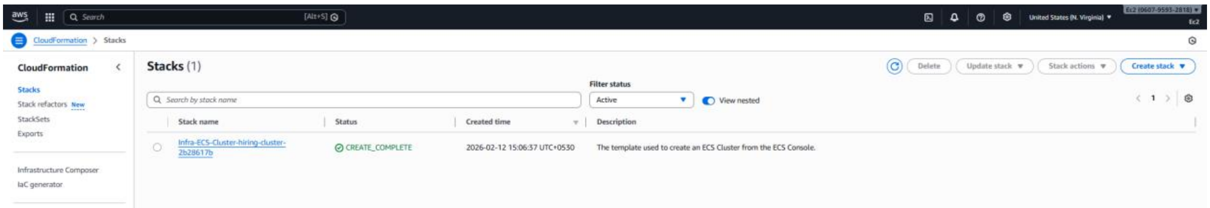


✅ PHASE 2: Confirm EC2 Instances Registered in ECS

Go to:
ECS → Clusters → hiring-cluster → Infrastructure

Check:

- EC2 instances should show as **ACTIVE**
- Container instances should show registered



If it shows 0 instances, wait 2-3 mins.

Yes bro. From Step 2 onward (cluster already creating), do this exactly:

✅ STEP 2: Verify EC2 Instances Joined ECS Cluster

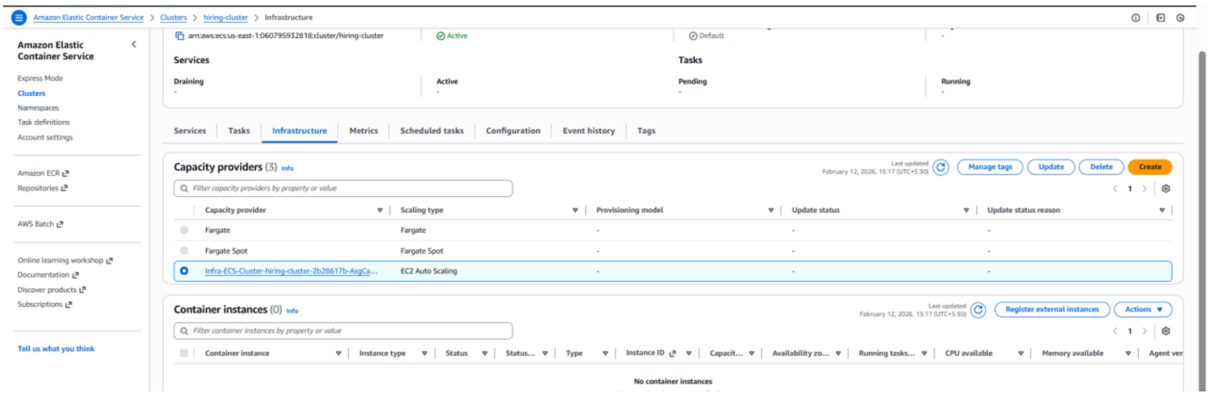
Go to:

ECS → Clusters → your-cluster → Infrastructure

Check:

- Container instances = 2 (or more)
- Status = ACTIVE

If 0, wait 2-3 mins.



✅ STEP 3: Create Application Load Balancer (ALB)

Go to:

EC2 → Load Balancers → Create Load Balancer → Application Load Balancer

Settings:

- Name: hiring-alb
- Scheme: Internet-facing
- Listener: HTTP 80
- VPC: Default VPC
- Subnets: select 2 subnets (2 AZ)

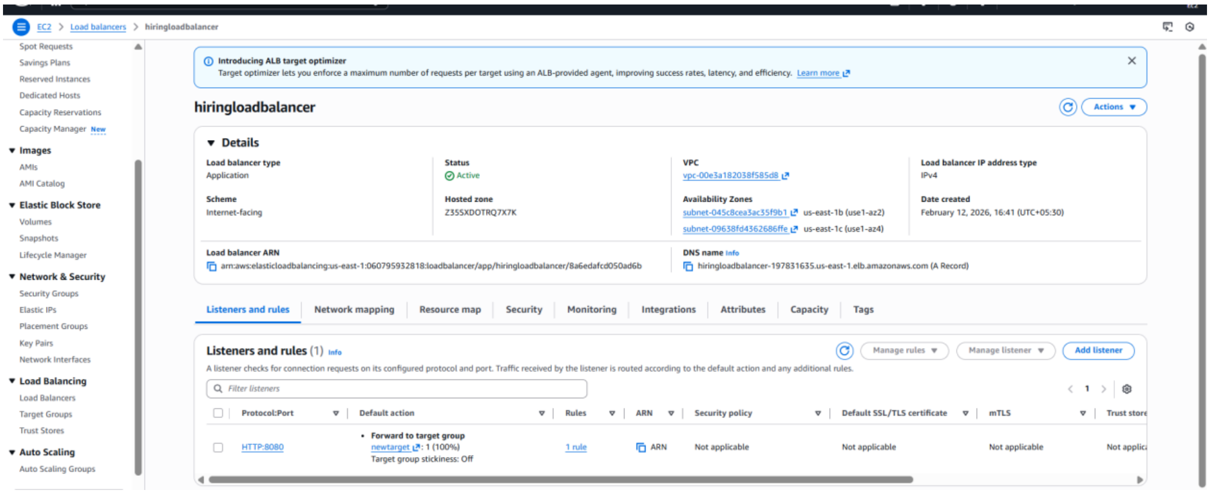
Create Security Group for ALB:

Name: alb-sg

Inbound rule:

- HTTP 80 from 0.0.0.0/0

Create ALB.



✅ STEP 4: Create Target Group

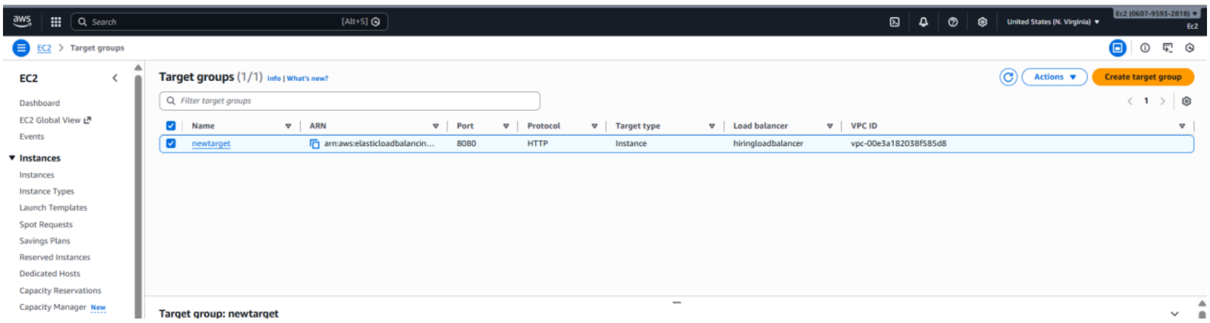
Go to:

EC2 → Target Groups → Create target group

Settings:

- Target type: ✅ Instance
- Protocol: HTTP
- Port: 80
- Name: hiring-tg
- Health check path: /

Create target group.



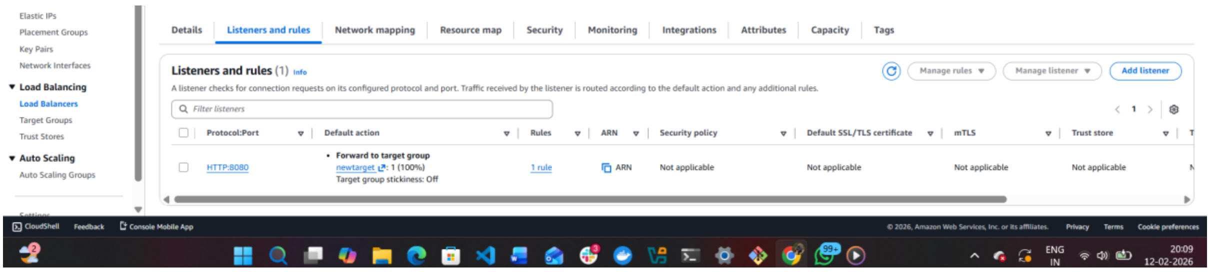
✅ STEP 5: Attach Target Group to ALB Listener

Go to:

EC2 → Load Balancers → ECRloadbalancer → Listeners → HTTP:80

Edit rule → Forward to:
✅ hiring-tg

Save.



✅ STEP 6: Update ECS EC2 Security Group (Very Important)

Go to:

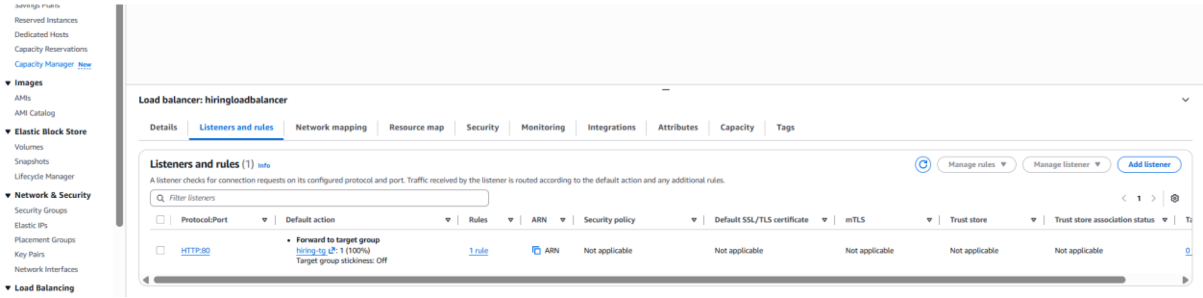
EC2 → Instances → select ECS instance → Security → open Security Group

Edit inbound rules and add:

Allow from ALB SG:

- HTTP 80 ➔ source: alb-sg
- TCP 32768-65535 ➔ source: alb-sg

(These ports are needed for dynamic port mapping)



✅ STEP 7: Create Task Definition (Using Given DockerHub Image)

Go to:

ECS ➔ Task Definitions ➔ Create new task definition

Choose:
✅ EC2

Task definition name:
hiring-task

Container:

Container name: hiring-app

Image:

sabair0509/hiring-app:works

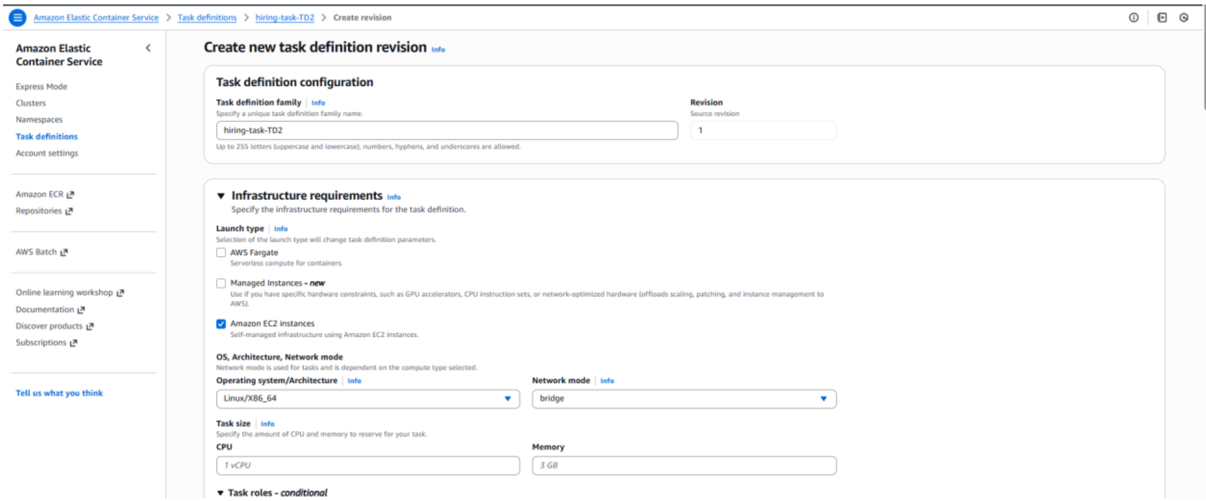
Port mapping:

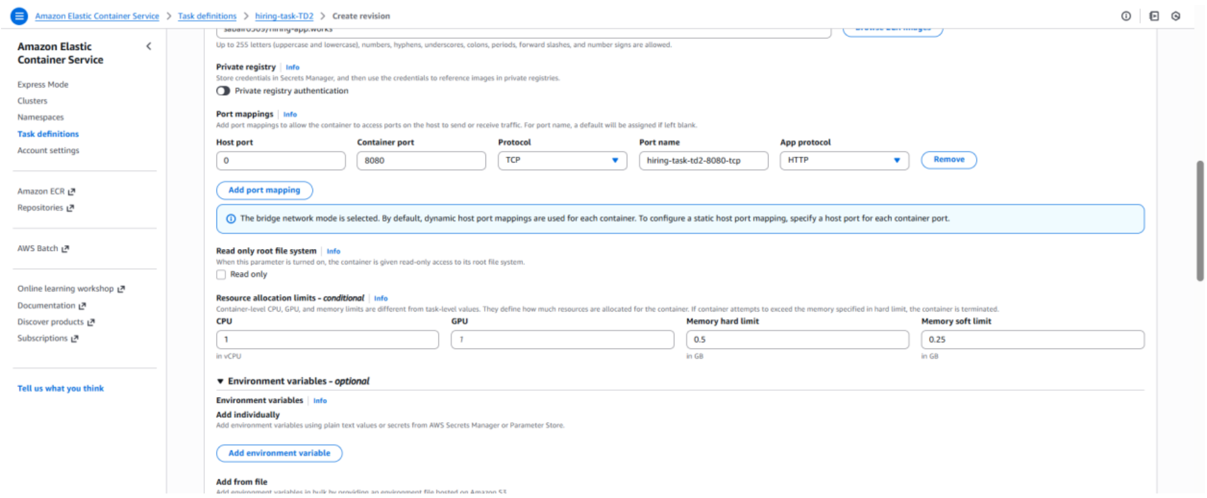
- Container port: 8080
- Host port: ✅ 0 (dynamic port mapping)

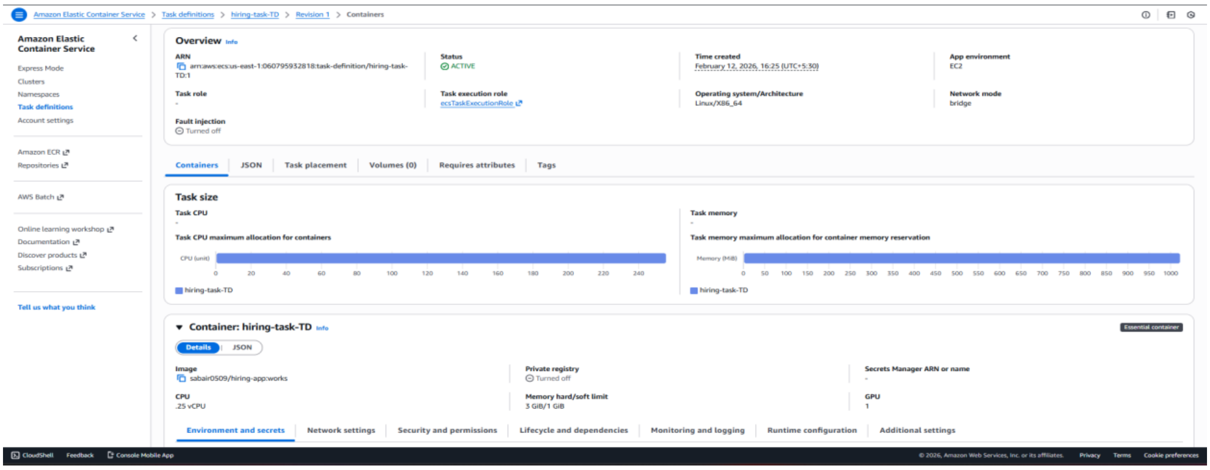
Memory reservation:
512

CPU:
256

Create task definition.







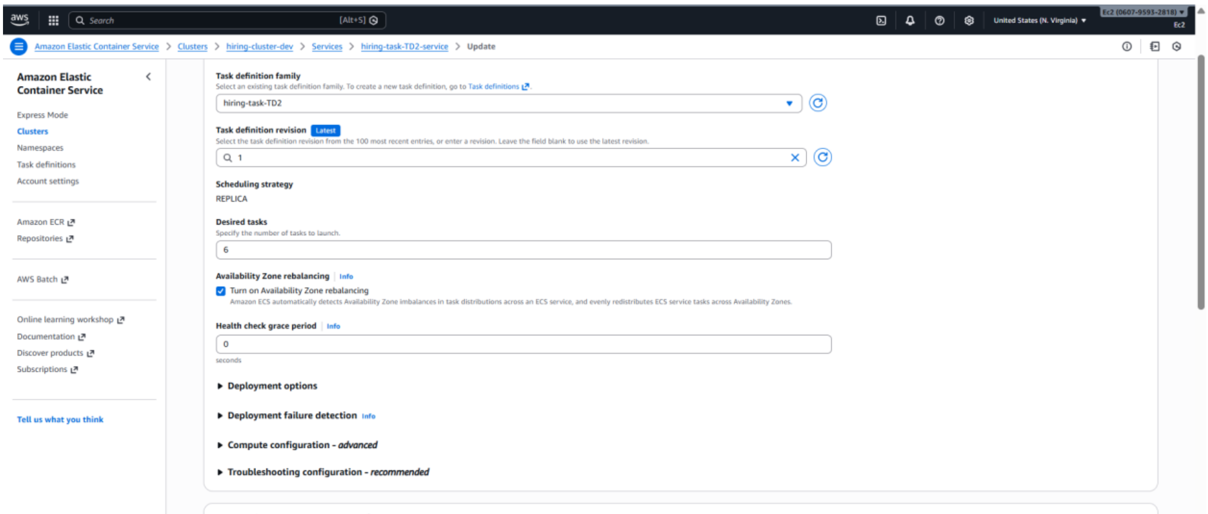
✅ STEP 8: Create ECS Service (Maintain 6 Tasks Always)

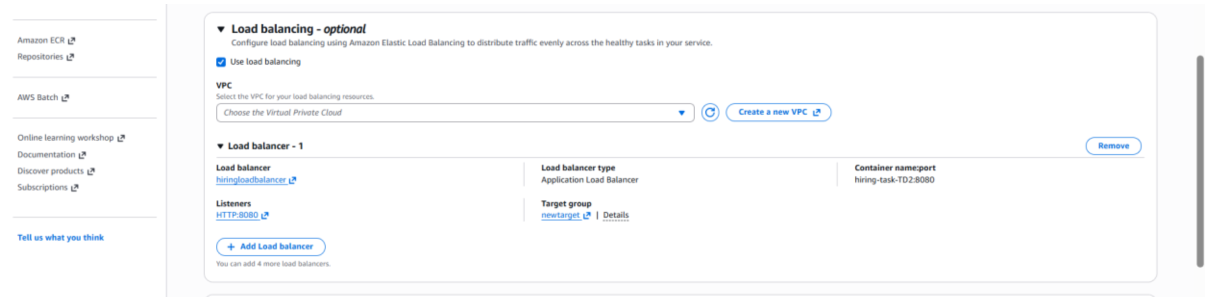
Go to:

ECS → Clusters → your-cluster → Services → Create

Settings:

Launch type: EC2
Service name: hiring-service
Task definition: hiring-task
Desired tasks: ✅ 6





✅ STEP 9: Configure Load Balancer in Service

In Load balancing section:

Enable:
✅ Application Load Balancer

Select:

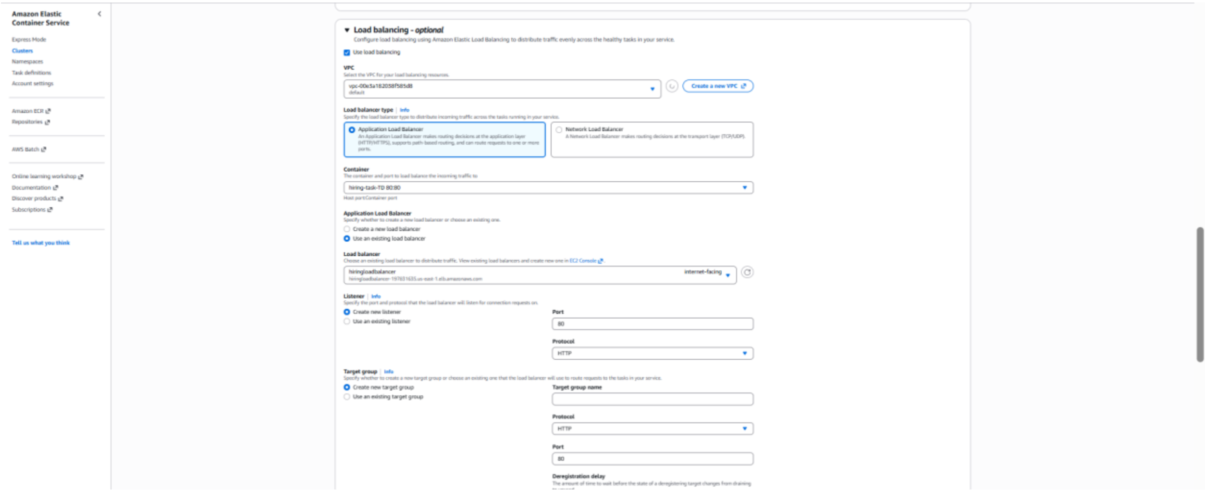
- Load balancer: hiring-alb
- Listener: HTTP 80

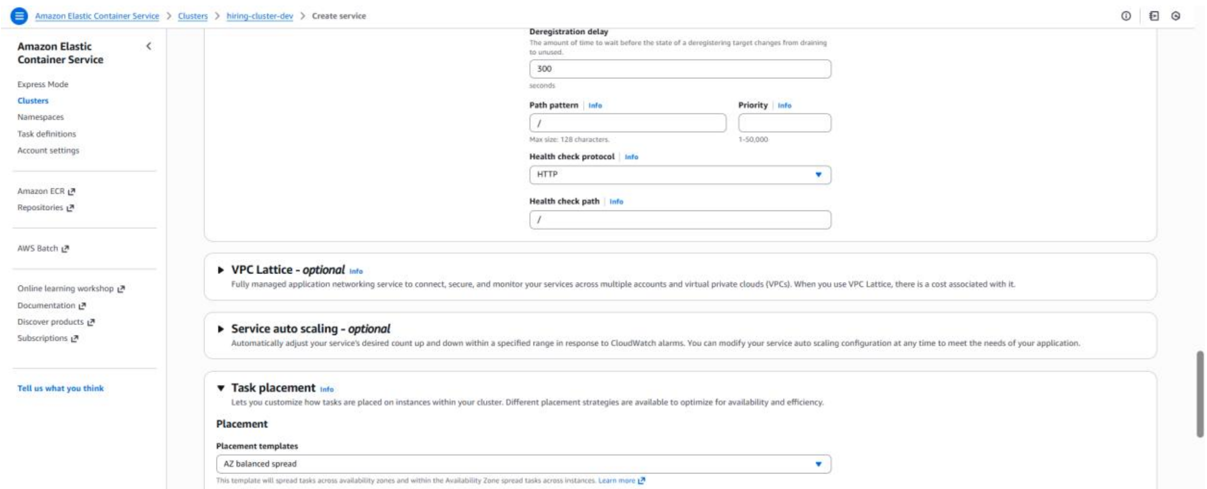
- Target group: hiring-tg

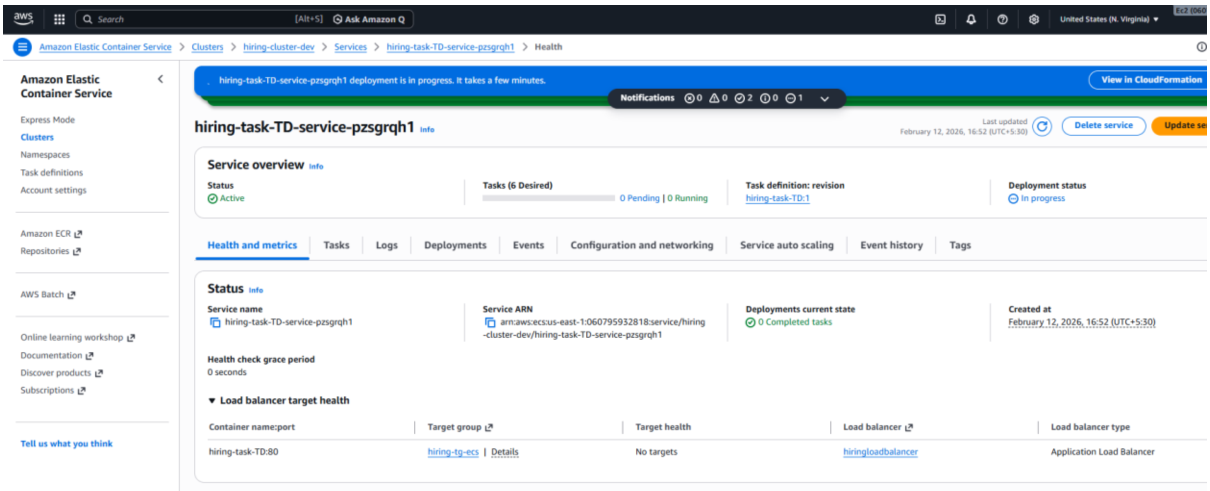
Container:

- hiring-app : 8080

Create service.







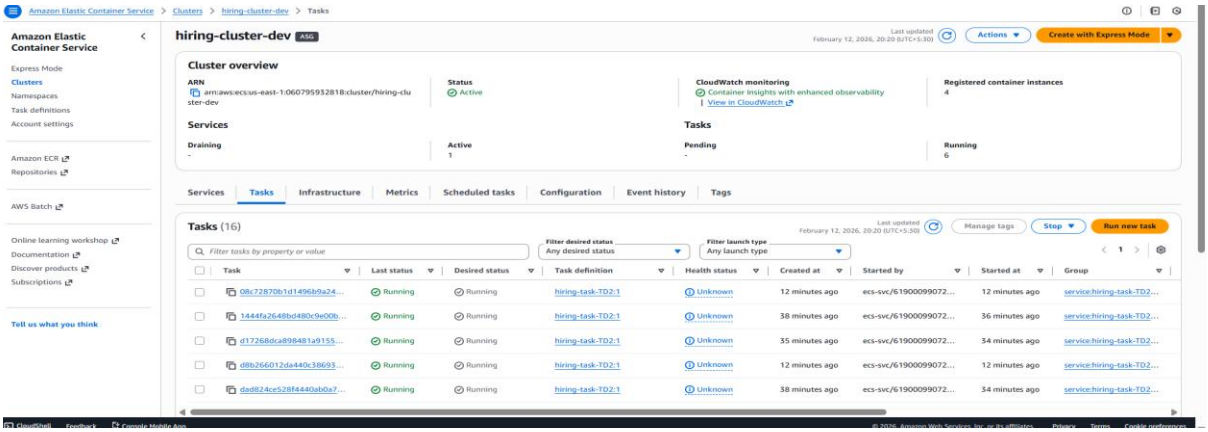
✅ STEP 10: Validate Tasks Running

Go to:

ECS → Cluster → Services → hiring-service → Tasks

You must see:
✅ 6 tasks running

If only 2-3 run:
➡️ Your EC2 capacity is low.
Increase Auto Scaling desired instances to 3 or change instance type to t3.medium.



✅ STEP 11: Validate Dynamic Port Mapping

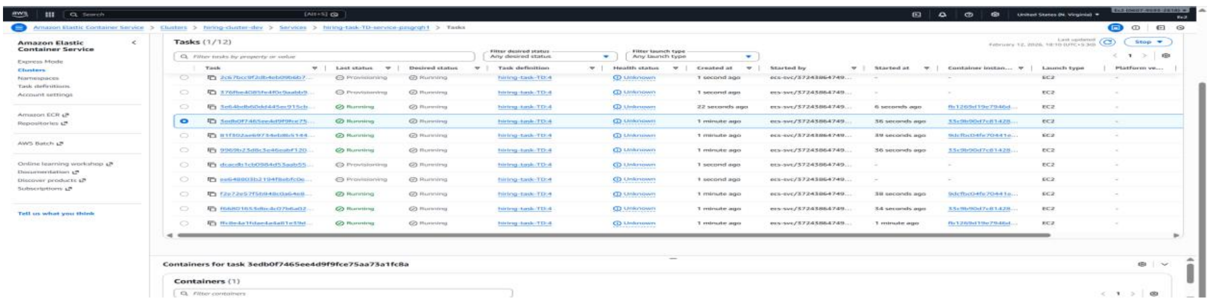
Go to:

EC2 → Target Groups → hiring-tg → Targets

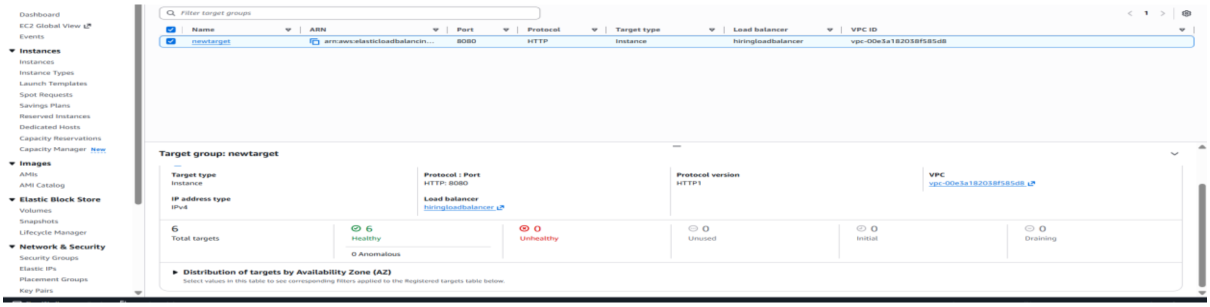
You should see registered targets like:

- i-xxxx : 32768
- i-xxxx : 32771

This confirms:
✅ dynamic port mapping working



→target group



✅ STEP 12: Final Test (ALB DNS)

Go to:

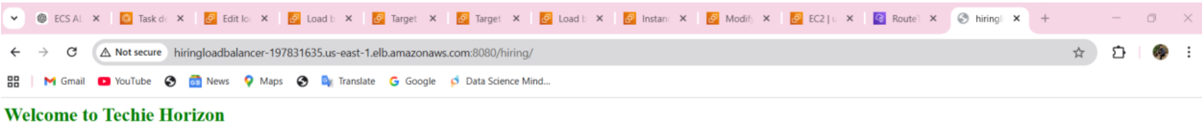
EC2 → Load Balancers → hiring-alb → DNS name

Open DNS in browser:

Example:

http://hiring-alb-xxxxx.us-east-1.elb.amazonaws.com

I should see the application running.



✅ FINAL CHECKLIST (Your Task Completed)

✅ Cluster EC2 in Multi-AZ
✅ ALB working
✅ Target group healthy
✅ Dynamic port mapping (host port 0)
✅ Service running 6 tasks always
✅ App accessible through ALB DNS